

산업 결함 Active Learning 후보 신호의 단계적 타당성 평가

Validity-gated empirical evaluation and cost-containment workflow

B Validity Gate Pipeline

A 초기 가설

VLM Consistency · DINO Diversity
Detector Uncertainty

↓
유용한 선택 → detector 성능 → label 비용 절감
점선 = 기각된 superiority 경로

G1
Signal validity

grounding
error ranking
transform confound

G2
Target discovery

rare/anomaly yield
enrichment@budget

G3
Composition safety

coverage · entropy
HHI · rare safety

G4
Acquisition reproducibility

new acquisition seed
fixed vs new set

G5
Learning utility

mAP50-95 · recall
rare AP · AULC

G6
Operational validity

top-budget enrichment
human/deployment value

Selection PASS는 성공 예측이 아니라, learning utility를 측정할 수 있는 1회 bounded detector screen 권한이다.

PASS → 다음 검증 단계

FAIL / NOT IDENTIFIABLE → STOP · 확장 금지 · claim 축소

Downstream FAIL → backbone 확장 금지 · final 잠금

C 데이터셋별 mechanism 사례

GC10-DET

Discovery
+2.720 rare images

Composition
proxy diversity ↑

Downstream
q20 overall mAP 일부 ↑
rare AP ↓ · K40 utility FAIL

Diversification without
reliable learning utility

MPDD

Discovery
+6.245 anomalies

Composition
capture-day ↑
source confound 67.3%

Downstream
downstream not tested

Source-confounded
diversification

VisA

Discovery
+14.480 anomalies

Composition
category -4.110
HHI +0.286025

Downstream
downstream not authorized

Category collapse

D 자동으로 성립하지 않은 연결

- × VLM consistency ≠ grounding
- × Global coverage ≠ local utility
- × Discovery gain ≠ composition safety
- × Train-seed stability ≠ acquisition generalization
- × Selection PASS ≠ detector success
- × Error AUROC ≠ budget enrichment
- × Metadata ≠ independent production pool

E 최종 산출 가치

Scientific output

8개 failure mechanism
discovery-safety-utility separation
claim-boundary enforcement

Cost containment

D2R 15 + K40 30
≥45 planned model runs stopped
locked final actual use = 0

RETROSPECTIVE EMPIRICAL EVALUATION
+ COST-CONTAINMENT WORKFLOW

PREDICTIVE POLICY: NOT IDENTIFIABLE

후보 신호의 부분적 성공을 detector utility로 곧바로 해석하지 않고, 각 번역 단계를 검증한 뒤 다음 고비용 단계의 수행 권한을 제한한다.